LECTURE L21-22 · MODULE 3 · CO3

Encoding strategies

Week 11 | Slides: `DATA209_Module_3_Data_Quality_Engineering.pptx`, pages 31–37 | Laboratory: **P21-22 (Assignment 2)**

TOPICS COVERED

- One-hot, label and target encoding
- Handling imbalanced and multi-class encoding
- Dummy variables and the dummy variable trap

Why encoding is a modelling decision

Most algorithms require numeric input, and the encoding you choose decides what the model can learn. Assigning 1, 2, 3 to unordered categories invents an order *and* a spacing that do not exist; the model will treat category 3 as "more" than 1 and as equidistant from 2.

For genuinely ordinal data, an ordered integer mapping is correct. For nominal data with few levels, one-hot is the safe default.

The schemes

SCHEME	WHAT IT DOES	USE WHEN	RISK
Label	one integer per level	ordinal data, or tree models	invents order on nominal data
Ordinal (mapped)	integers in a stated order	true ordinal, order known	assumes equal spacing
One-hot	one binary column per level	nominal, low cardinality	column explosion; dummy trap
Binary / hashing	compact numeric encoding	high cardinality	collisions; loses interpretability
Frequency	replace level with its share	high cardinality, frequency meaningful	unrelated levels share a value
Target / mean	replace level with mean target	high cardinality, predictive levels	**leakage**

Start with one-hot. Move to something cleverer only when cardinality forces you, and then be explicit about the risk.

Two traps

The dummy variable trap. One-hot on k levels produces k perfectly collinear columns — any one is exactly one minus the sum of the others. Linear models cannot invert the resulting matrix, so coefficients become unstable or undefined. Fix: drop one level as the reference (`drop_first=True`). Trees and regularised models are unaffected, but the redundancy remains.

Target encoding leakage. Replacing a level with its mean target uses the answer to build the feature. Computed on the full dataset, every row sees its own target: validation looks excellent and live performance collapses. Fix: compute the mapping inside cross-validation folds, on training data only, and add smoothing so rare levels are pulled toward the global mean.

> If a feature seems implausibly predictive, suspect leakage before celebrating.

Harder cases

High cardinality — group rare levels into "Other" by a stated frequency threshold before encoding. Keep the threshold as a parameter and report it; "levels under 1% of rows" is reproducible. **Unseen levels** at prediction time break a naive encoder — set `handle_unknown='ignore'`.

Imbalance changes evaluation, not encoding, but note it here. At 84.5% / 15.5%, accuracy is not an honest metric: report precision, recall, F1 and PR-AUC, and choose by which error costs more. Stratify every split and every cross-validation fold. Resampling — oversampling, undersampling, SMOTE — belongs *inside the training fold only*; applied before the split it duplicates minority rows across train and test, which is leakage. Class weights are often simpler and safer.

Multi-class targets — check that every class survives the split and that rare classes have enough rows to learn from.

KEY TERMS

`One-hot encoding` `Label encoding` `Ordinal encoding` `Target encoding` `Dummy variable trap` `Reference level` `High cardinality` `handle_unknown` `Stratified split` `SMOTE`

COMMON ERRORS TO AVOID

- Label-encoding a nominal variable for a linear model
- One-hot encoding without dropping a reference level in a linear model
- Computing target encoding on the full dataset
- Resampling before the train/test split

READINGS — ALL FREE TO ACCESS

1. scikit-learn — preprocessing and encoding
   `https://scikit-learn.org/stable/modules/preprocessing.html`
2. scikit-learn — common pitfalls and data leakage
   `https://scikit-learn.org/stable/common_pitfalls.html`
3. Kuhn & Johnson — Feature Engineering and Selection
   `http://www.feat.engineering/`

TEST YOURSELF

1. Why does one-hot on k levels break an unpenalised linear model?
2. Describe the correct way to compute target encoding.
3. A categorical column has 4,000 levels. Give two viable strategies.
4. Why is resampling before the split a form of leakage?